Simple model of volcanic gas dispersion

Connor - **Victor volcano workshop**

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1 Introduction

This suggested workshop project involves estimating hazards due to volcano degassing. The idea is to develop a notebook that calculates expected gas concentration as some location, given conditions at the volcanic vent and given specific atmospheric conditions. In the following, some background is provided, followed by a simplified mathematical approach for quantifying hazards. This approach is widely used for describing any sort of atmospheric contaminant transport, but includes the unique thermal and flux conditions at a volcanic vent.

2 Background on gas dispersion

Volcanic gas is released through the ground, from vents, and from volcanic products before, during, and following volcanic eruptions. Volcanic gases create hazards primarily because they can occur at very high emission rates. The composition of volcanic gas varies widely, but gases that can create significant hazard impacts include SO_2, CO_2, HCl, HF, CO and H_2S.

Hazards due to volcanic gas emissions might arise due to proximity to an actively degassing volcanic system. The goal of modeling volcanic gas emissions for hazard assessment is to determine an appropriate offset distance from an actively degassing volcanic system, beyond which volcanic gases will not accumulate in sufficient concentrations to cause harm.

3 Modeling gas concentration

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Volcanic gas concentration in the atmosphere, down wind of an actively degassing volcanic system, is modeled using the advection-diffusion equation, shown here in 2D:

$$\frac{\partial C_{i,j}}{\partial t} + w_x \frac{\partial C_{i,j}}{\partial x} + w_z \frac{\partial C_{i,j}}{\partial z} = K \frac{\partial^2 C_{i,j}}{\partial x^2} + K \frac{\partial^2 C_{i,j}}{\partial z^2} + \Phi \tag{1}$$

where, x and z are spatial coordinates expressed in meters; $C_{i,j}$ is the concentration of gas; w_x is the horizontal wind velocity, which may be a function of elevation ($\mathrm{m\,s^{-1}}$); w_z is the vertical plume velocity, which may be a function of Brigg's buoyancy flux parameter and the horizontal wind speed, K is the diffusion coefficient for gas in the atmosphere; Φ is the change in gas concentration at the source with time, t.

A challenge is that volcanic gas emissions are long-lived associated with the opening of a new vent. Atmospheric conditions will likely vary considerably during the period of gas emissions. Consequently models could be developed to consider the likely worst case conditions to cause unacceptably high gas concentrations at a given downwind receptor location at any time during the period of active degassing. That is, the hazard assessment for gas emissions considers the offset distance from the vent, beyond which volcanic gases will not accumulate in sufficient concentrations to prevent an operational response regardless of the atmospheric conditions or level of flux from the volcanic system.

Table 1: Pasquill stability class coefficients for Turner's equation.

Stability class	I_z	J_z	K_z
A	6.035	2.1097	0.2270
B	4.694	1.0629	0.0136
C	4.110	0.9201	-0.0020
D	3.414	0.7371	-0.0316
E	3.057	0.6794	-0.0450
F	2.621	0.6564	-0.0540
	I_y	J_y	K_y
A	5.357	0.8828	-0.0076
B	5.058	0.9024	-0.0096
C	4.651	0.9181	-0.0076
D	4.230	0.9222	-0.0087
E	3.922	0.9222	-0.0064
F	3.533	0.9191	-0.0070

To make this estimate we simplify the model to an analytical solution using Turner's dispersion equation and Brigg's equation for bent-over plumes, with an analytical solution to the advection-diffusion equation.

Turner's dispersion equation accounts for the change in dispersion in horizontal (y) and vertical (z) directions as a function of downwind distance, x (km):

$$\sigma_z = \exp(I_z + J_z \log_{10} x + K_z (\log_{10} x)^2) \tag{2}$$

$$\sigma_y = \exp(I_y + J_y \log_{10} x + K_y (\log_{10} x)^2) \tag{3}$$

The coefficient values ($I_z, J_z, K_z, I_y, J_y, K_y$) are found empirically for Pasquill atmospheric stability classes (Table 1).

Then the ground concentration is found:

$$C = \frac{Q}{u_x \sigma_z \sigma_y \pi} \exp\left[\frac{-y^2}{2\sigma_y^2}\right] \exp\left[\frac{-h^2}{2\sigma_z^2}\right] \tag{4}$$

where the ground concentration is in microgram/m^3, Q is the emission rate at the volcanic vent (microgram/s), u_x is the horizontal wind speed (m/s), σ_z and σ_y are the vertical and crosswind diffusion coefficients, respectively; y is the distance crosswind from the plume center-line ($y = 0$ on the plume center-line), and h is the height of the plume center-line.

Given a volumetric flow rate at the vent, V (Nm^3/hr), one can calculate the gas flow rate (kg-mol/s):

$$Q_g = \frac{V}{22.41} \times \frac{1}{3600} \tag{5}$$

For example, for SO_2 concentration in the vent gas, C_{SO_2}, yields an SO_2 emission rate in microgram/s (there are 10^9 micrograms in one kilogram):

$$Q_{SO_2} = Q_g C_{SO_2} \times 10^{-6} \times MW_{SO_2} \times 10^9 \tag{6}$$

where MW_{SO_2} is the molecular weight of $SO_2 = 64$

The gas velocity (m/s) at the vent is:

$$u_e = \frac{V}{\pi r^2} \left[\frac{T_{gas} + 273}{273}\right] \times \frac{1}{3600} \tag{7}$$

Table 2: Wind velocity is calculated as a function of height for a given stability class. This table gives the exponent, n used in equation 12 as a function of Pasquill stability class.

Stability class	n
A	0.10
B	0.15
C	0.20
D	0.25
E	0.25
F	0.30

Brigg's equation for bent-over plumes. The plume buoyancy parameter is:

$$F = g u_e r^2 \frac{T_{gas} - T_{air}}{T_{gas}} \tag{8}$$

where g is gravity, u_e is the exit velocity of the gas at the vent, r is the hydraulic radius of the conduit (equal to fracture aperture for a long fissure), T_{gas} is the temperature of the gas (K), T_{air} is the air temperature (K).

The stability parameter, s is:

$$s = \frac{g}{T_{air}} \cdot \text{abs}\left[\frac{d\theta}{dz}\right] \tag{9}$$

where $d\theta/dz$ is the temperature gradient.

Downwind rise distance until the plume center-line no longer changes height is:

$$x_f = 119 F^{0.4} \text{ , if } F \geq 55 \tag{10}$$

$$x_f = 49 F^{0.625} \text{ otherwise} \tag{11}$$

The wind speed is found as a function of height above the vent for a given Pasquill stability class:

$$u_x(z) = u_s \left[\frac{h_z}{h_s}\right]^n \tag{12}$$

where u_s is the wind speed measured at a height h_s above the surface, and h_z is the height above the surface at which $u_x(z)$ is calculated. n depends on Pasquill stability class (Table 2).

Brigg's equation describes the plume center-line height as a function of atmospheric stability. If the stability class is A, B, C, or D, corresponding to relatively unstable atmospheres:

$$\Delta h = 1.6 F^{1/3} x^{2/3} u_x^{-1} \text{ , if } x < x_f \tag{13}$$

$$\Delta h = 1.6 F^{1/3} x_f^{2/3} u_x^{-1} \text{ otherwise} \tag{14}$$

If the stability class is E or F, corresponding to stable atmospheres with positive lapse rate (temperature

decreases as a function of height):

$$S = \frac{1.84 u_x}{\sqrt{s}} \tag{15}$$

$$\Delta h = 1.6 F^{1/3} x^{2/3} u_x^{-1} \text{ , if } S \geq x_f \text{ and } x < x_f \tag{16}$$

$$\Delta h = 1.6 F^{1/3} x_f^{2/3} u_x^{-1} \text{ , if } S \geq x_f \text{ and } x \geq x_f \tag{17}$$

$$\Delta h = 1.6 F^{1/3} x^{2/3} u_x^{-1} \text{ , if } S < x_f \text{ and } x < S \tag{18}$$

$$\Delta h = 2.4 \left[\frac{F}{u_x s} \right]^{1/3} \text{ , otherwise} \tag{19}$$

Once the SO_2 emission rate, Q_{SO_2} (microgram/s) and vent flow rate V are determined, the procedure is to:

1. Determine the gas exit velocity u_e at the vent, given the vent flow rate (Nm^3/s), the vent radius (m), and the vent gas temperature (K).
2. Calculate the stability parameter, s
3. Calculate the Brigg's buoyancy parameter
4. Find the distance downwind that the plume reaches its final height (x_f)
5. Find the wind velocity at the height of the vent
6. Find the height of the plume center-line Δh above receptor located at x using Brigg's equations, adjusting for the vent height
7. Find the wind velocity at the center-line height above the receptor location, x
8. Calculate the ground concentration

Maximum downwind gas concentration along the plume center line at ground level was estimated using this approach, equations and algorithm.

4 What to do?

Consider implements code in a notebook to:

1. implement the equations for a given gas flux and vent conditions, for a point located some distance from the volcano along the plume centerline
2. plot the expected SO_2 gas concentration as a function of distance downrange of the volcano for the same vent conditions but different windspeeds or diffeent atmospheric stability class
3. write a code to monte carlo sample vent conditions for expected atmospheric conditions at a given distance from the vent
4. consider how to incorporate local atmospheric data